

1) Base and Limit Address Translation

```
#include <stdio.h>
int main() {
    int base, limit, logical;
    printf("Enter base, limit and logical address: ");
    scanf("%d%d%d", &base, &limit, &logical);
    if (logical >= 0 && logical < limit)
        printf("Physical Address = %d", base + logical);
    else
        printf("Invalid Address");

    return 0;
}
```

OUTPUT:

Enter base, limit and logical address: 5

5

5

Invalid Address

=== Code Execution Successful ===

2)Paging Memory Management

```
#include <stdio.h>
int main() {
    int page, offset, size, frame;
    printf("Enter page size, page number, frame number, offset: ");
    scanf("%d%d%d%d", &size, &page, &frame, &offset);

    if (offset < size)
        printf("Physical Address = %d", frame * size + offset);
    else
        printf("Invalid Offset");

    return 0;
}
```

OUTPUT:

Enter page size, page number, frame number, offset: 100 2 5 20

Physical Address = 520

3. Page Table Simulation

```
#include <stdio.h>

int main() {
    int page, offset, size, frame[10];

    printf("Enter page size: ");
    scanf("%d", &size);

    printf("Enter frame numbers for pages 0,1,2: ");
    scanf("%d%d%d", &frame[0], &frame[1], &frame[2]);

    printf("Enter page number and offset: ");
    scanf("%d%d", &page, &offset);

    if (page >= 0 && page < 3 && offset < size)
        printf("Page %d -> Frame %d\nPhysical Address = %d",
            page, frame[page], frame[page] * size + offset);
    else
        printf("Invalid Address");

    return 0;
}
```

OUTPUT:

Enter page size: 100

```
Enter frames for pages 0, 1, 2: 3 5 2
Enter page number and offset: 1 40
Page 1 -> Frame 5
Physical Address = 540
```

4. TLB Simulation

```
#include <stdio.h>
```

```
int main() {
    float hit, tlb = 10, mem = 100, eat;

    printf("Enter TLB hit ratio: ");
    scanf("%f", &hit);

    eat = hit * (tlb + mem) +
          (1 - hit) * (tlb + 2 * mem);

    printf("Without TLB = %.2f ns\n", 2 * mem);
    printf("With TLB = %.2f ns", eat);

    return 0;
}
```

OUTPUT:

```
Enter TLB hit ratio: 0.8
Without TLB = 200.00 ns
With TLB = 130.00 ns
```

5. Memory Protection

```
#include <stdio.h>
```

```
int main() {
    int base, limit, logical;

    printf("Enter base, limit and logical address: ");
    scanf("%d%d%d", &base, &limit, &logical);

    if (logical >= 0 && logical < limit)
        printf("Access Allowed\nPhysical Address = %d",
               base + logical);
    else
        printf("Unauthorized Access");

    return 0;
}
```

OUTPUT:

```
Enter base, limit and logical address: 2000 500 450
Access Allowed
Physical Address = 2450
```